

Crystallization control of 2.0-eV bandgap perovskites for all-perovskite triple-junction solar cells

Received: 10 December 2025

Accepted: 6 August 2026

Published online: 10 September 2026

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Ultrawide-bandgap (UWBG) perovskites are essential for monolithic all-perovskite triple-junction solar cells. However, their performance is limited by severe surface wrinkling, halide heterogeneity and large open-circuit voltage deficits originating from high Br–I ratios. Here we develop a surface reconstruction and halide homogenization strategy that simultaneously regulates crystallization and eliminates stress-induced morphological disorder in 2.0-eV bandgap perovskites. Through synergistic solvent and additive engineering, involving solvent-induced surface reconstruction and transient Cl[−] incorporation from oleylammonium chloride, surface wrinkling is suppressed, and Br/I crystallization is synchronized with promoted homogeneous halide distribution. The resulting UWBG perovskite films show uniform surface potential, suppressed non-radiative recombination, improved carrier mobility and an open-circuit voltage of 1.46 V. Integrating this UWBG subcell with optimized 1.6- and 1.22-eV absorbers yields monolithic all-perovskite triple-junction solar cells with 30.1% power conversion efficiency (certified 29.3%) and robust operational stability, providing a promising route towards >35% efficient all-perovskite multijunction photovoltaics.

Wide-bandgap (WBG) perovskite solar cells (PSCs) are the preferred top absorbers for state-of-the-art perovskite–silicon and all-perovskite multijunction architectures, enabling efficiencies beyond the Shockley–Queisser (S–Q) limit for single junctions^{1–3}. Optical and device modelling has suggested that monolithic all-perovskite tandems using ~1.8-/1.2-eV absorbers can theoretically reach ~33.8% power conversion efficiency (PCE), while triple-junction solar cells (TJSCs) combining ~2.0-, 1.6- and 1.2-eV perovskites can achieve efficiencies of up to 36.6%, comparable to those of state-of-the-art III–V multijunction technologies^{3–5}. While rapid progress has pushed monolithic all-perovskite tandems to 30.1% PCE very recently⁶, the development of all-perovskite triple-junction

devices remains limited, with only a few demonstrations to date achieving efficiencies of 27.4% for monolithic all-perovskite TJSCs, far below their theoretical potential⁷.

The major bottleneck for TJSCs is the ultrawide-bandgap (UWBG, >1.9 eV) top subcell. Achieving such bandgaps requires high Br–I ratios, but Br-rich compositions introduce severe crystallization challenges^{8–10}. Rapid antisolvent extraction promotes fast crystallization of the Br-rich top region, forming a relatively rigid surface capping layer, while the underlying I-rich region remains viscous and amorphous. During subsequent solvent evaporation and densification, shrinkage of the bottom layer generates compressive mechanical

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mismatch with the rigid top layer, leading to buckling-driven wrinkle formation and large-area morphological roughness^{11–13}. In parallel, the high Br content accelerates halide heterogeneity and the formation of I-rich domains, causing non-uniform local chemical potential, carrier trapping and large open-circuit voltage (V_{oc}) deficits that frequently exceed ~200 mV, which is far larger than the <50 mV typical of middle-bandgap (MBG) perovskites (optical bandgap (E_g), 1.5–1.6 eV)¹⁴. This morphological and compositional heterogeneity restricts both efficiency and long-term reliability in UWBG subcells and therefore limits the performance of all-perovskite TJSCs. While several additive engineering strategies (for example, diammonium salts) have been explored to reduce halide segregation, they often fail to suppress stress-induced wrinkling or light-induced instabilities^{1,2,8,9,15}.

In this work, we introduce a surface reconstruction and halide homogenization (SRH) strategy that simultaneously controls surface morphology and halide crystallization kinetics in UWBG perovskites. A solvent treatment simultaneously reconstructs the wrinkled surface and promotes rapid Br–I exchange in the wet film, while oleylammonium chloride (OAmCl) participates in the intermediate phase crystallization by incorporating Cl^- into early nuclei, synchronizing Br/I incorporation and suppressing halide segregation during growth. This cooperative mechanism eliminates stress-driven wrinkling, minimizes local compositional gradients and yields UWBG films with uniform surface potential, low trap density, improved crystallinity and enhanced carrier mobility. Consequently, we achieve a V_{oc} of 1.46 V, corresponding to ~87% of the S-Q limit for 2.0-eV bandgap absorbers. Monolithic all-perovskite TJSCs comprising 2.0-, 1.6- and 1.22-eV absorbers deliver a PCE of 30.1% (29.3% certified) with a V_{oc} of 3.52 V. The TJSCs retain 90% of their initial PCE after ~600 h of maximum power point (MPP) tracking under continuous 1-sun illumination and 240 h under damp-heat stress (85 °C, 85% relative humidity) and show only 9% degradation after 200 thermal cycles between –40 °C and 85 °C. These results show that resolving the UWBG bottleneck through coupled surface and crystallization control unlocks the path towards highly efficient and stable all-perovskite multijunction photovoltaics.

SRH strategy

By controlling the I–Br ratio to 1:2, we demonstrate a UWBG perovskite material $FA_{0.7}Cs_{0.3}PbI_2Br$ with an E_g of ~2.0 eV, which is ideal for front cell applications in triple-junction tandem photovoltaics as discussed above^{3,5}. Consequently, perovskite films fabricated via conventional anti-solvent quenching develop pronounced wrinkle-like surface features (Supplementary Fig. 1a). The SRH strategy is illustrated in Fig. 1a. In this approach, immediately after the conventional antisolvent dripping step, an as-cast wet film is treated with a solubility-engineered solvent mixture that enables controlled surface reconstruction (Fig. 1a and Supplementary Fig. 2). Specifically, a mixture of hexafluoro-isopropanol (HFIP) and *N,N*-dimethylformamide (DMF) (30:1, v/v) (denoted as HD) acts as the surface reconstruction solvent. DMF dissolves a partial surface layer, whereas HFIP, as a strongly hydrogen-bond-donating but weakly coordinating solvent¹⁶, limits excessive lattice dissolution, thereby enabling partial reconstruction of the perovskite surface. Consequently, this mixed solvent system enables controlled surface reflow, suppressing over-etching and the formation of undesirable 2D phases associated with PbI_2 exposure¹⁷. This synergistic interaction results in a self-limited etching process that smooths the surface without compromising the underlying lattice (Supplementary Figs. 3 and 4).

Optical microscopy confirms that this treatment substantially suppresses wrinkle formation and improves surface flatness in the as-cast wet films (Fig. 1b). Nevertheless, surface wrinkling is largely alleviated in the wet films, and surface and cross-sectional scanning electron microscopy (SEM), together with optical microscopy, reveals the reappearance of wrinkles upon annealing (Fig. 1c,d and Supplementary Fig. 5). This indicates that the morphological instability is not fully resolved

and is likely driven by a kinetically induced Br–I nucleation imbalance during recrystallization. To address the Br–I nucleation imbalance and suppress the morphological disorder in annealed films, OAmCl is incorporated into the HD solvent (denoted as HDCl), where it acts as a nucleation balance species by temporarily incorporating Cl^- into early perovskite nuclei, where it has been reported to have a bridging effect between Br and I (ref. 18). As a result, HDCl-treated films show substantially improved thickness uniformity after thermal annealing (Fig. 1c and Supplementary Fig. 5), indicating effective surface planarization and facilitated halide redistribution.

The time-of-flight secondary ion mass spectrometry (TOF-SIMS) depth profile shows a trace but detectable OAm⁺ signal ($C_{18}H_{38}N^+$), which is primarily localized near the top surface rather than distributed throughout the bulk (Supplementary Fig. 6). This indicates that OAm⁺ does not incorporate into the perovskite lattice but instead accumulates at the film surface after annealing. To further clarify the role of OAm⁺, we systematically compared alkylammonium chlorides with different chain lengths. The OAmCl-treated devices show the highest device performance and longest carrier lifetime, outperforming those based on shorter chain ligands (see Supplementary Note 1 and Supplementary Figs. 7 and 8 for details). This can be attributed to the preferential surface accumulation of long-chain OAm⁺, which effectively passivates surface defects and suppresses interfacial recombination. Compared to shorter alkyl chains, the hydrophobic nature and larger molecular size of OAm⁺ enable stronger surface coverage and more effective defect passivation, accounting for its superior performance. These results are consistent with previous reports¹⁹.

The X-ray diffraction (XRD) patterns of all as-cast wet films show distinct diffraction double peaks from the (100) crystal planes, with peak splitting observed at approximately 14.8°/15.1° (Fig. 1e). The peak at 15.1° corresponds to the diffraction of $FA_{0.7}Cs_{0.3}PbBr_3$ films (Supplementary Fig. 9), while the lower angle peak corresponds to a mixed Br–I phase. The SRH strategy promotes a redistribution of anions, thereby enabling the formation of a more compositionally homogeneous Br–I mixed phase after annealing (Fig. 1f). The full width at half maximum of the (100) peak was further calculated for the annealed films. The full width at half maximum of the HD- and HDCl-treated film is substantially narrower than that of the control (0.149° versus 0.171°), confirming that the SRH strategy promotes a homogeneous Br–I mixed phase after annealing. In addition, the HD-treated perovskite films reveal exposed PbI_2 , and the introduction of OAmCl does not lead to the formation of a 2D phase (Supplementary Note 2 and Supplementary Figs. 10 and 11). This is attributed to the strong interaction between OAmCl and HFIP, which disrupts the OAmCl– PbI_2 interaction, as further confirmed by ¹H and ¹⁹F nuclear magnetic resonance spectroscopy (see Supplementary Note 3 and Supplementary Figs. 12 and 13 for details).

A mixture of isopropanol (IPA) and DMF (denoted as ID) was also investigated for comparison, with the corresponding morphological data provided in Supplementary Figs. 14 and 15. Both HDCl- and IDCl-treated films show slightly reduced average thicknesses but significantly narrower thickness distributions compared to the control sample. This indicates that the SRH process enables effective surface planarization without substantial net material loss. Instead, it induces surface material redistribution, likely through partial dissolution of the surface layer followed by reorganization, leading to a more uniform and compact film (Supplementary Note 4). Although IDCl-treated perovskite films show a noticeable reduction in surface wrinkling, a distinct 2D phase emerges after thermal annealing. This phase formation is likely to result in higher interfacial resistance and poor device performance (Supplementary Fig. 16). In addition, the effects of both HD solvent ratio and OAmCl concentration on film morphology and device performance have been systematically investigated, and the results have been added in Supplementary Figs. 17–20.

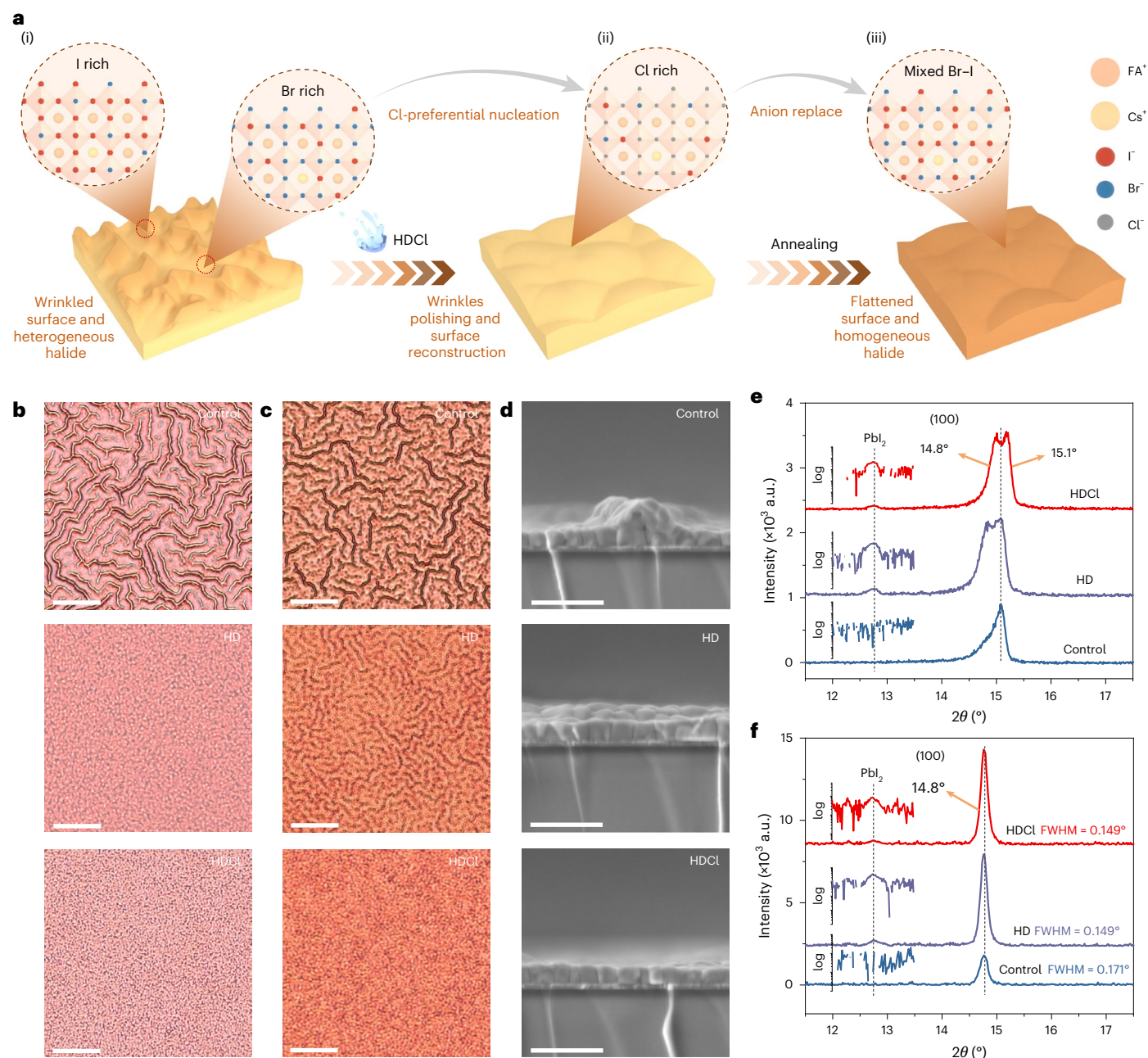


Fig. 1 | Morphology and structure of UWBG perovskite with SRH strategy. **a**, Schematic illustrations of morphological and halide homogenization processes. **b, c**, Optical microscopy images of the perovskite films before (**b**) and after (**c**) annealing. Scale bar, 50 μm . **d**, Cross-sectional SEM images of the

perovskite films. Scale bar, 1 μm . **e, f**, XRD patterns of the perovskite films before (**e**) and after (**f**) annealing. a.u., arbitrary units. XRD profiles (log-scaled intensity) in the 2θ range of 12° – 13.5° for the inset.

In addition, XRD patterns of as-cast wet films reveal a distinct wide-angle shift of the (100) reflections in the HDCl film before annealing (Fig. 1e), indicating the presence of Cl-rich intermediates in the wet films¹⁸. The absence of this shift after annealing, combined with the disappearance of the Cl signal in X-ray photoelectron spectroscopy (XPS) (Supplementary Fig. 21), suggests that the majority of Cl⁻ is removed during crystal growth and largely replaced by Br⁻ and I⁻. To further validate Cl⁻ participation, we precipitated intermediate phases by gradually adding anisole into the precursor solution (Supplementary Fig. 22). The control solution produced orange APbBr₃-rich precipitates, while OAmCl-containing solutions yielded yellow Cl-containing intermediates^{20,21}. The XRD analysis of dried powders from antisolvent extraction shows a pronounced wide-angle shift in diffraction peaks for the OAmCl sample (Supplementary Fig. 23),

and the XPS analysis confirms the presence of Cl-rich intermediate signatures (Supplementary Fig. 21). Combined with the XRD evolution observed in wet and annealed films, these results show that Cl⁻ participates exclusively at the intermediate stage, is removed from the lattice upon annealing and is ultimately replaced by Br⁻ and I⁻ during crystal growth. This transient Cl⁻ templating effect accelerates Br⁻/I⁻ homogenization, promotes more uniform domain formation and substantially suppresses surface wrinkling in the final UWBG perovskite films¹¹.

Homogeneity of UWBG perovskite films

Optical microscopy and SEM images in Fig. 1 show pronounced surface wrinkling in conventionally processed films (control). We have performed atomic force microscopy (AFM) measurements on both the control and SRH-treated perovskite films (Supplementary Fig. 24).

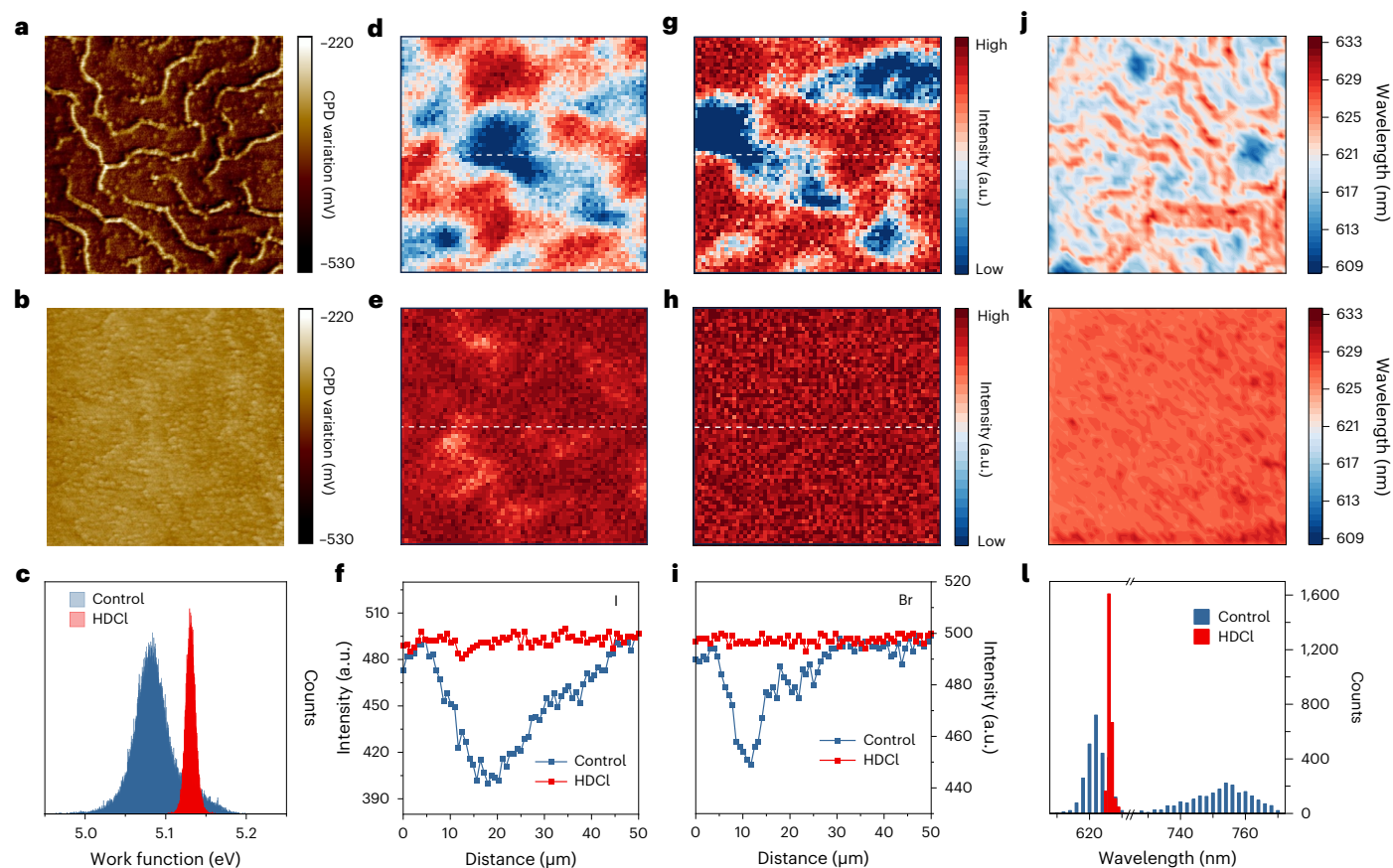


Fig. 2 | Homogeneity of UWBG perovskite films. **a, b**, KPFM characterization of surface potential distribution in control (**a**) and HDCl (**b**) films. **c**, Statistical distribution of work functions for perovskite films, derived from **a** and **b**. **d, e**, TOF-SIMS intensity maps of I for the control and HDCl films. **f**, Corresponding TOF-SIMS line-scan intensity profiles of I for control and HDCl films. **g, h**, TOF-SIMS

intensity maps of Br for the control and HDCl films. **i**, Corresponding TOF-SIMS line-scan intensity profiles of Br for control and HDCl films. The blue and red lines in **f** and **i** show the paths of the line scans in the mapping figures. **j–l**, PL wavelength mapping and distributions of control (**j**) and HDCl-treated (**k**) films. Note that all map images share the same imaging area of $50 \times 50 \mu\text{m}^2$.

AFM analysis reveals root-mean-square roughness values of 110.0 nm for the control film and 63.9 nm for the SRH-treated film. This result is consistent with earlier observations from optical microscopy and SEM, which showed suppressed wrinkle formation and improved morphological uniformity. Kelvin probe force microscopy (KPFM) further reveals that these morphological irregularities coincide with substantial compositional heterogeneity, producing spatial variations in the local work function (Fig. 2a–c and Supplementary Figs. 25 and 26). The contact potential difference map of the control film shows strong variations across the film and a broad work-function distribution of 5.0–5.2 eV. Such variations are indicative of local changes in I/Br stoichiometry. Consistent with previous reports, halide inhomogeneity leads to spatial bandgap fluctuations, where I-rich regions possess narrower bandgaps and Br-rich regions show wider bandgaps^{11,22}. By sharp contrast, the HDCl film shows a smooth and continuous contact potential difference landscape, and its work function histogram collapses into a sharp peak centred at 5.13 eV, reflecting a laterally uniform halide distribution and strongly suppressed bandgap variations. This improved compositional uniformity reduces surface energy barriers for carrier transport and enables more homogeneous charge extraction^{23,24}.

To probe compositional heterogeneity within the UWBG perovskite film, we performed TOF-SIMS mapping and depth profiling. As revealed by the mapping results (Fig. 2d–i), the control film shows clear spatial segregation of Br-rich and I-rich domains, confirming that halides crystallize at different rates within the rough, wrinkled films. In comparison, the HDCl film shows continuous and highly uniform Br and I distributions through the film, indicating that the SRH

approach effectively suppresses phase partitioning. It should be noted that SEM images indicate wrinkle features with widths of $\sim 1\text{--}2 \mu\text{m}$, which are notably smaller than the lateral scale of halide heterogeneity observed in TOF-SIMS ($\sim 30 \mu\text{m}$). This apparent discrepancy arises because wrinkling and halide heterogeneity are coupled manifestations of the same underlying heterogeneous crystallization process but occur at different length scales. Specifically, the wrinkling features observed by microscopy represent local morphological responses to stress and compositional inhomogeneity, whereas TOF-SIMS maps capture broader Br- and I-rich domains formed during crystallization. Therefore, individual wrinkles do not directly correspond to halide domains of the same size. Instead, multiple local wrinkles can form within larger regions of compositional inhomogeneity.

Wavelength-dependent photoluminescence (PL) mapping further verifies these observations (Fig. 2j–l and Supplementary Fig. 27). The control film shows pronounced spatial variations in its emission peak position, where two distinct peaks are observed. The emergence of new, lower-energy PL peaks ($>700 \text{ nm}$) was attributed to I-rich phase separation, reflecting substantial local bandgap fluctuations induced by halide heterogeneity. This is consistent with the work function variations observed in the KPFM measurements. By contrast, the HDCl film shows a highly uniform PL peak position with a markedly narrower statistical distribution, confirming its improved halide compositional uniformity (Fig. 2k, l). In addition, the HDCl film shows a substantially brighter and more spatially homogeneous PL intensity compared to the control, suggesting improved crystallinity and reduced non-radiative recombination in the UWBG perovskite film (Supplementary Fig. 28).

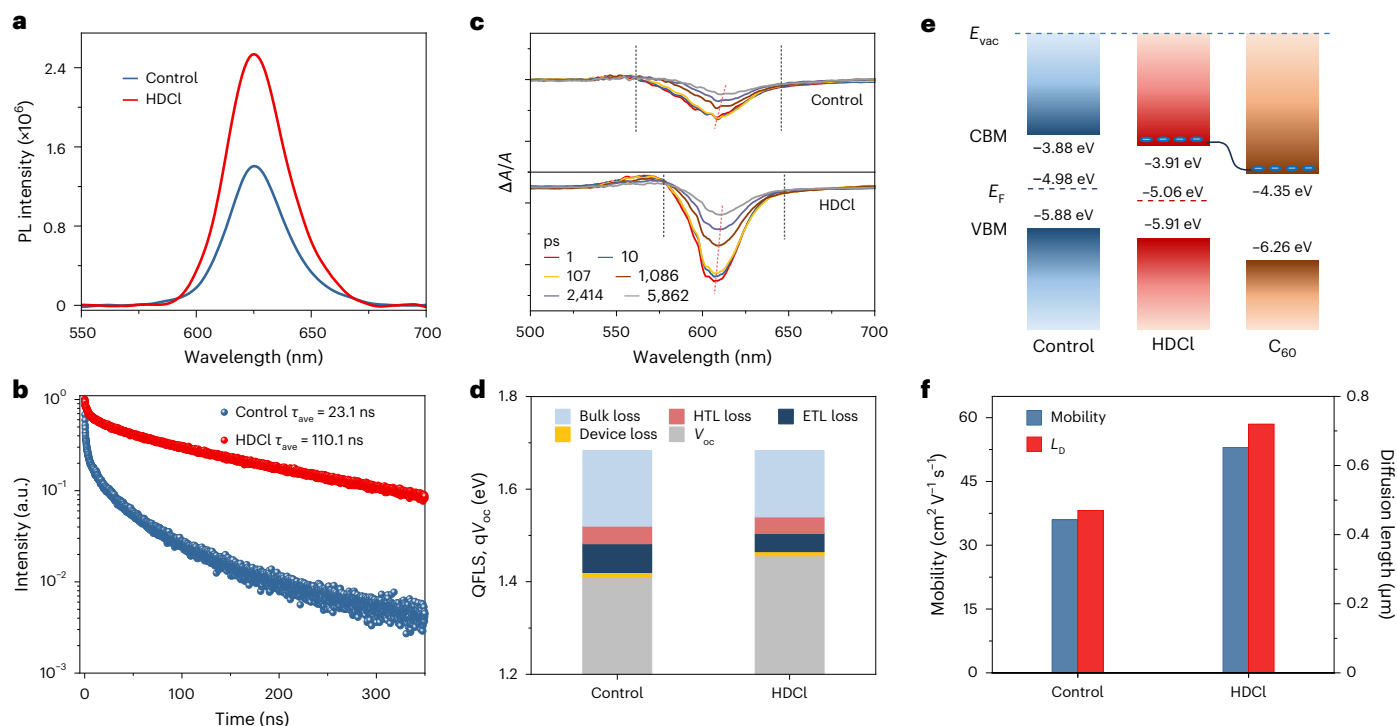


Fig. 3 | Defects and charge carrier dynamics. **a, b**, Steady-state PL (**a**) and TRPL (**b**) curves of control and HDCl films on glasses. **c**, Ultrafast TA spectra of control and HDCl perovskite films under 405-nm pump excitation, where $\Delta A/A$ is the pump-induced relative change in absorbance, with ΔA corresponding to the change in absorbance and A to the probe absorbance. Time delays are denoted

in picoseconds (ps). **d**, QFLS and V_{oc} values of control and HDCl perovskite films deposited on glass (neat), with C₆₀ on top and on ITO/NiO. **e**, Energy band diagrams at the interface between perovskite films and C₆₀, showing the vacuum level (E_{vac}), Fermi level (E_F), conduction band minimum (CBM) and valence band maximum (VBM). **f**, Mobility and diffusion length of the control and HDCl perovskite films.

Defects and carrier dynamics of UWBG perovskite films

We evaluate the defect passivation effect of UWBG perovskite films using PL and time-resolved photoluminescence (TRPL) measurements. It was observed that the HDCl perovskite film showed approximately 1.8 times higher PL intensity than the control film (Fig. 3a). This indicates that SRH treatment reduces the non-radiative recombination in the UWBG perovskite film. We further carried out the TRPL decays of the control and HDCl films (Fig. 3b). When the film was excited from the glass side, the charge recombination lifetime (τ) dramatically decreased from 110.1 ns (HDCl) to 23.1 ns (control), and this observed trend was reproducible across multiple measurements (Supplementary Fig. 29), indicating effective suppression of carrier trapping throughout the UWBG perovskite layer. We further investigated the charge-generation and recombination dynamics in the UWBG perovskite films using ultrafast transient absorption spectroscopy^{25,26} (TA; Fig. 3c and Supplementary Fig. 30). At a probe wavelength of 610 nm, both films show a negative signal corresponding to ground-state bleaching (GSB). The HDCl perovskite film shows a narrower and less shifted GSB peak compared to the control, consistent with improved halide compositional homogeneity as discussed above. In addition, the HDCl film shows a noticeably slower GSB decay, indicating a prolonged carrier lifetime and suggesting that the SRH process effectively suppresses trap-assisted recombination (Supplementary Fig. 31).

To quantify non-radiative recombination losses in our p-i-n devices, we performed photoluminescence quantum yield measurements to calculate the quasi-Fermi level splitting (QFLS) of the control and HDCl samples²⁷ (Fig. 3d and Supplementary Table 1). Compared to the control film with voltage loss (E_{loss}) of 163 meV, the HDCl-treated film showed a reduced E_{loss} of 143 meV, indicating the effectiveness of our SRH strategy in passivating defects within the UWBG perovskite neat film. Analysis of interfacial losses reveals that the hole transport

layer (HTL) contributes relatively minor losses (~38 meV), whereas the electron transport layer (ETL) interface introduces markedly larger losses (62 meV), identifying the perovskite/ETL interface as a major performance limiting factor. Consistently higher QFLS values in the HDCl sample further confirm improved bulk and interfacial properties enabled by the SRH strategy. Ultraviolet photoemission spectroscopy measurements show a reduced conduction band offset between the perovskite and C₆₀ layer in the HDCl sample, leading to improved energy-level alignment and more efficient electron extraction (Supplementary Fig. 32 and Fig. 3e).

Femtosecond-resolved optical-pump terahertz-probe (OPTP) spectroscopy was used to probe carrier transport²⁸. Charge-carrier mobilities (sum of hole and electron mobilities) were extracted using the Drude-Smith model²⁹, with detailed fitting procedures provided in Supplementary Note 5 (Supplementary Figs. 33–36 and Table 2). The HDCl film shows a notably higher mobility ($53 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$) compared to the control ($36 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$), which contributes directly to its extended carrier lifetime (Fig. 3f). Consequently, the charge-carrier diffusion lengths (L_D), calculated from the OPTP decay dynamics, increase from 0.47 μm in the control film to 0.72 μm in the HDCl film, reflecting substantial improvements in long-range carrier transport. To identify the mechanistic origins of the enhanced diffusion length and extended lifetime, we quantified the trap density of states ($tDOS$) and their energetic distribution using thermal admittance spectroscopy³⁰ (Supplementary Fig. 37). The HDCl device shows a markedly lower $tDOS$ in the shallow-trap regime (0.20–0.35 eV), indicating suppressed trap formation and reduced non-radiative recombination.

Performance of single-junction UWBG cells

The inverted UWBG single-junction solar cells with the architecture glass/indium tin oxide (ITO)/NiO_x/Me-4PACz/FA_{0.7}Cs_{0.3}PbIBr₂/C₆₀/ALD-SnO₂/Cu were fabricated (Fig. 4a). All devices incorporated

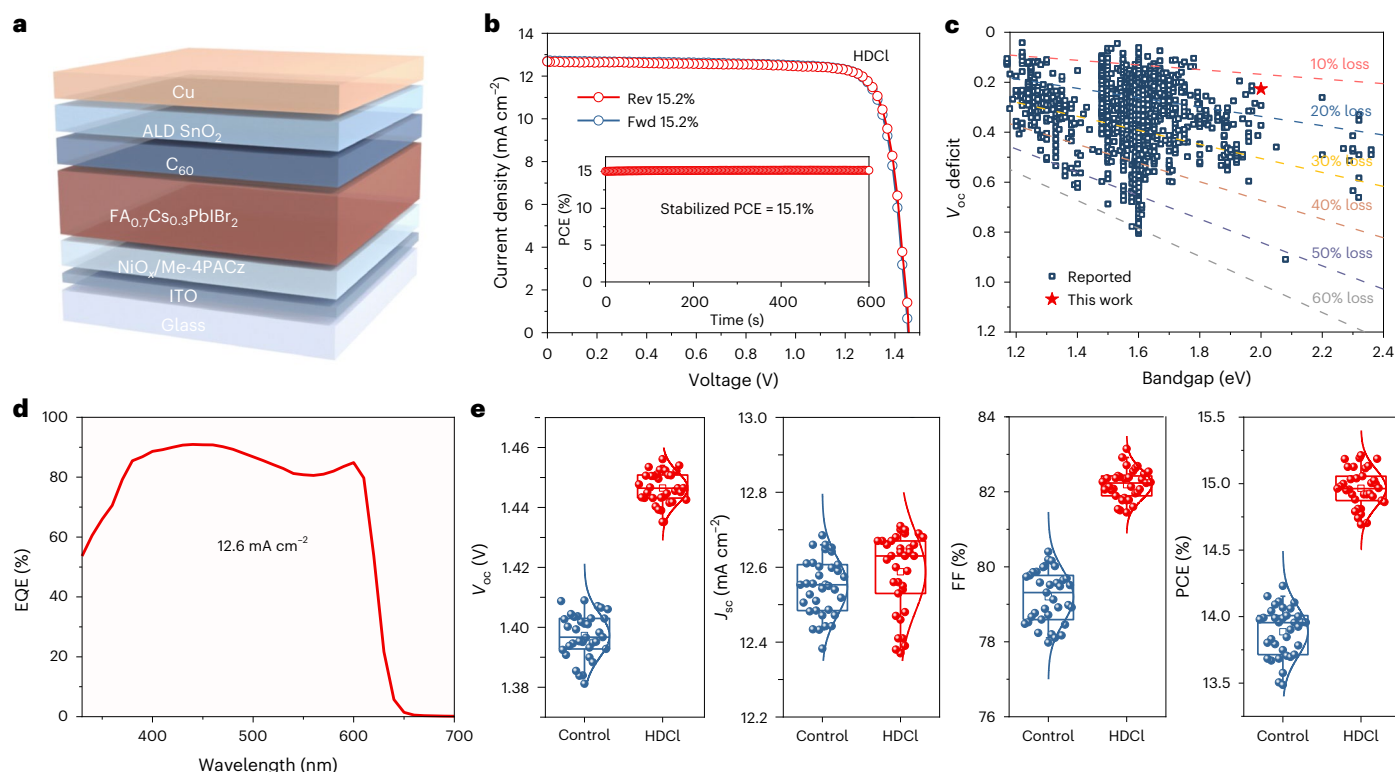


Fig. 4 | The performance of single-junction UWBG devices. **a**, Schematic of the device structure. **b**, The J - V curves of the best-performing HDCl UWBG solar cells under reverse (Rev) and forward (Fwd) scanning directions, with an inset showing the stabilized PCE. **c**, A summary of the V_{oc} deficit at different E_g values for state-of-the-art PSCs. The literature cell data were acquired from an open-access perovskite database³⁷. In the plot, the S-Q limit (dashed line), the V_{oc} deficit of this work (solid star) and open source data (open brackets) are shown.

d, The EQE spectra of the best-performing HDCl UWBG solar cell. **e**, Comparison of photovoltaic performance between control and HDCl UWBG solar cells processed in the same runs (35 devices for each type, biological replicates). Scatters represent raw data; boxplots show the interquartile range with the median as the centre line; whiskers indicate the 5th–95th percentiles. Numerical data are presented as mean $\pm$ standard deviation (s.d.).

previously reported surface-passivation steps in which sequential interface engineering was applied to optimize the perovskite/ C_{60} interface and suppress interfacial recombination⁶. Representative cross-sectional SEM images are shown in Supplementary Fig. 38, and full fabrication details are provided in the Methods. Devices processed via our strategy showed reduced surface wrinkling and enhanced halide homogeneity, and achieved a champion PCE of 15.2%, substantially higher than that of the control (14.2%), with notable enhancements in V_{oc} (1.46 V versus 1.40 V) and fill factor (FF) (82.3% versus 80.4%) (Fig. 4b, Supplementary Fig. 39 and Supplementary Table 3). Figure 4c summarizes V_{oc} deficits as a function of E_g for state-of-the-art PSCs. Narrow-bandgap (NBG, $E_g = 1.1$ – 1.3 eV), MBG ($E_g = 1.5$ – 1.6 eV) and WBG ($E_g = 1.7$ – 1.8 eV) perovskites generally show relatively small voltage deficits, with the best devices losing only ~10% relative to their respective S-Q limits. By contrast, UWBG perovskites suffer from substantially larger V_{oc} deficits, with many devices showing >20% losses, largely attributable to high trap densities and photoinduced halide segregation. Our champion UWBG cell achieves a V_{oc} deficit of only ~13% relative to its S-Q limit, showing its strong potential for high-voltage triple-junction integration.

Capacitance–voltage measurements (10 kHz, dark) further reveal that this V_{oc} enhancement originates primarily from an increase in the built-in potential, which rises from 1.36 V in the control device to 1.41 V in the HDCl device³¹ (Supplementary Fig. 40). Under continuous operation at the MPP for 600 s, the champion device maintained a stabilized PCE of 15.1% (Fig. 4b, inset). The external quantum efficiency (EQE) spectrum of the champion device yields an integrated photocurrent of 12.6 mA cm^{-2} , in agreement with the current density–voltage (J - V) characteristics (Fig. 4d). The devices also show excellent reproducibility,

with an average PCE of 15.0% across 35 independently fabricated cells (Fig. 4e and Supplementary Table 4).

To gain deeper insight into the origin of the improved V_{oc} in HDCl devices, we first performed transient photovoltage and electrochemical impedance spectroscopy (EIS) measurements^{32,33}. The τ of the HDCl device increased by nearly 20-fold relative to the control (2 μ s versus 0.1 μ s), consistent with the longer carrier lifetime and lower trap density observed in UWBG films (Supplementary Fig. 41). EIS measurements show that the HDCl device exhibits a recombination resistance (R_{rec}) of $6.8 \text{ M}\Omega \text{ cm}^{-2}$, nearly twice that of the control device ($3.5 \text{ M}\Omega \text{ cm}^{-2}$), indicating substantially suppressed recombination (Supplementary Fig. 42). This result is consistent with the light-intensity-dependent V_{oc} analysis, where the ideality factor decreases from 1.69 in the control device to 1.24 in the HDCl device³⁴ (Supplementary Fig. 43). Sensitive EQE spectroscopy further reveals a lower Urbach energy for the HDCl device ($E_u = 22.7 \text{ mV}$) compared to the control ($E_u = 26.9 \text{ mV}$), reflecting reduced band-edge disorder^{7,11} (Supplementary Fig. 44).

Performance of triple-junction devices

Leveraging the optimized UWBG subcell, we constructed all-perovskite triple-junction devices using the device architecture ITO/HTL/UWBG perovskite/ETL/ITO/MBG perovskite/ETL/Au/HTL/NBG perovskite/ETL/Cu (Methods). Optical simulations from previous studies indicate that achieving a champion efficiency in all-perovskite triple-junction architectures requires absorber bandgaps of approximately 2.0, 1.6 and 1.20 eV for the top, middle and bottom subcells, respectively. Guided by this design principle, we developed $\text{FA}_{0.8}\text{CS}_{0.2}\text{Pb}(\text{I}_{0.8}\text{Br}_{0.2})_3$ and $\text{FA}_{0.7}\text{MA}_{0.3}\text{Pb}_{0.5}\text{Sn}_{0.5}\text{I}_3$ perovskite absorbers with bandgaps of ~1.6 and ~1.22 eV

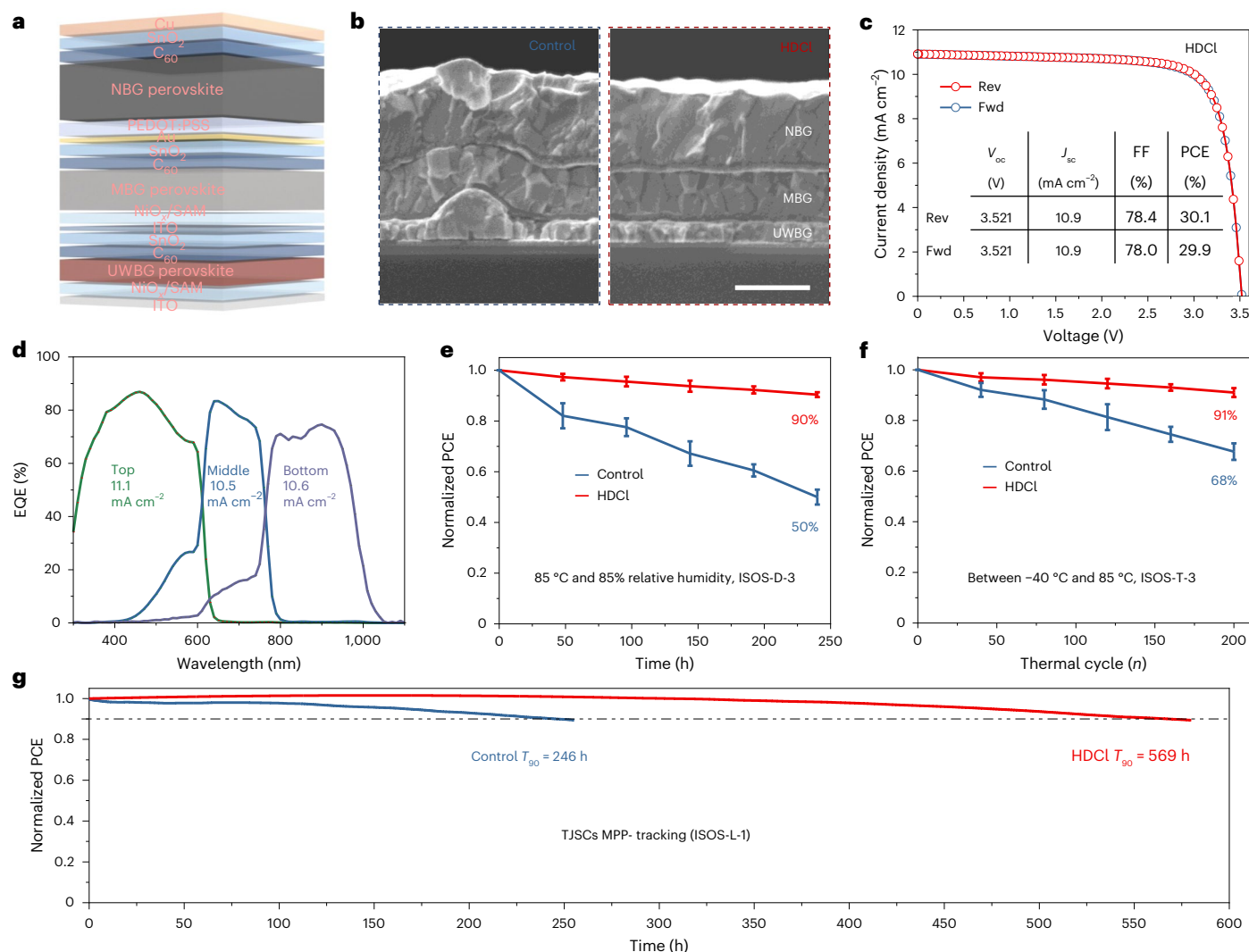


Fig. 5 | The performance of TJSCs. a, Schematic device structure of the all-perovskite TJSC. **b**, Cross-sectional SEM images of control and HDCl-based devices. Scale bar, 1 μm . **c, d**, The J - V (**c**) and EQE (**d**) curves of the champion all-perovskite TJSC. **e**, Encapsulated TJSCs aged under damp-heat conditions (85 °C, 85% relative humidity) and open-circuit conditions for 240 h. Solid lines show mean values, and error bars indicate mean $\pm$ s.d. The initial average PCEs for the control and HDCl-based devices ageing groups were 25.3 $\pm$ 2.43% and 27.9 $\pm$ 0.84%, respectively (n = 5, biological replicates). **f**, Encapsulated TJSCs

aged under thermal cycling conditions (between -40 °C and 85 °C). The coloured areas are error bars. Solid lines show mean values, and error bars indicate mean $\pm$ s.d. The initial average PCEs for the control and HDCl-based devices ageing groups were 24.9 $\pm$ 1.02% and 28.0 $\pm$ 0.79%, respectively (n = 5, biological replicates). **g**, Continuous MPP tracking of the encapsulated TJSCs under simulated AM 1.5 G illumination (100 mW cm^{-2} , LED simulator) in ambient air with a humidity of 30%–50%. The initial PCEs for the control and HDCl-based devices were 26.5% and 28.8%, respectively.

to serve as the middle and bottom subcells (Supplementary Figs. 45 and 46). The complete device architecture and interconnection layout are illustrated in Fig. 5a.

Cross-sectional SEM imaging of the HDCl-based triple-junction devices reveals uniform and well-controlled layer thicknesses, with the UWBG, MBG and NBG subcells measuring ~270 nm, ~580 nm and ~1,000 nm, respectively (Fig. 5b and Supplementary Fig. 47). By stark contrast, the control device shows pronounced surface irregularities across all-perovskite three-junction devices (Fig. 5b), largely originating from the severe wrinkling present in the UWBG subcell, and achieves a PCE of 28.5% (Supplementary Fig. 48). As a result of this improved structural uniformity, the HDCl-based triple-junction devices deliver markedly enhanced performance. The best-performing device achieves a PCE of 30.1%, with V_{oc} = 3.52 V, J_{sc} = 10.9 mA cm^{-2} and FF = 78.4%, marking the first monolithic all-perovskite TJSC to surpass the 30% efficiency threshold (Fig. 5c and Supplementary Table 5). The HDCl device shows an exceptionally small V_{oc} deficit of only 20 mV relative to the summed V_{oc} (3.54 V) of the constituent single-junction cells,

which is far lower than previously reported deficits exceeding 80 mV (Supplementary Table 6). This improvement primarily arises from the suppressed surface wrinkling and improved halide homogenization in the UWBG perovskite subcell enabled by our processing approach.

The integrated J_{sc} values from EQE measurements are 11.1, 10.5 and 10.6 mA cm^{-2} for the UWBG, MBG and NBG subcells, respectively (Fig. 5d), which is consistent with the J - V characteristics. The fabrication process also shows excellent reproducibility, where statistical analysis of 35 independently fabricated devices shows an average PCE of 29.6 $\pm$ 0.3% (Supplementary Fig. 49). Key performance metrics for all fabricated all-perovskite TJSCs are summarized in Table 1. Furthermore, the champion device was independently certified by the Shanghai Institute of Microsystem and Information Technology, yielding a certified PCE of 29.3% under reverse scan with minimal hysteresis (index = 1.5%). Full certification details are provided in Supplementary Fig. 50.

Beyond efficiency, stability is another critical challenge for all-perovskite triple-junction devices. We first evaluated the damp-heat stability of the triple-junction cells following the International Summit

Table 1 | Photovoltaic parameters of the control and HDCl-based TJSCs

Device	Scan direction	V_{oc} (V)	J_{sc} (mA cm ⁻²)	FF (%)	PCE (%)
Control	Reverse	3.460	10.8	76.2	28.5
	Forward	3.446	10.8	75.2	28.0
	Average	3.441±0.019	10.8±0.1	75.5±0.6	28.0±0.3
HDCl	Reverse	3.521	10.9	78.4	30.1
	Forward	3.521	10.9	78.0	29.9
	Average	3.515±0.009	10.8±0.1	77.8±0.4	29.6±0.3

on Organic Photovoltaic Stability (ISOS) dark storage protocol at 85 °C and 85% relative humidity (ISOS-D-3). As shown in Fig. 5e, the HDCl-based TJSCs retained 90% of their initial PCE, whereas the control devices showed a pronounced decay, maintaining only 50% of their original performance. We further investigated thermal cycling stability following the ISOS thermal cycling protocol, subjecting the devices to temperature cycles between -40 °C and 85 °C (ISOS-T-3). After 200 cycles, the HDCl-based TJSCs showed only 9% degradation, in contrast to a 32% loss observed in the control devices (Fig. 5f). Photostability was also evaluated under the ISOS light-soaking protocol at ambient (ISOS-L-1), where encapsulated TJSCs were operated in air at 25 °C under continuous 1-sun AM 1.5 G illumination with MPP tracking. The initial $J-V$ curves and the spectral irradiance of the LED simulator used for the light soaking tests have been included as Supplementary Figs. S1 and S2. The HDCl-based TJSCs retained over 90% of their initial PCE after 569 h, whereas the control devices reached T_{90} at only 246 h (Fig. 5g). These results show that the enhanced morphological and compositional homogeneity imparted by the HDCl processing strategy markedly strengthens the environmental robustness of all-perovskite TJSCs.

Conclusions

In summary, we identify severe surface wrinkling and halide heterogeneity in high-Br/I UWBG perovskite films as the principal bottlenecks limiting all-perovskite triple-junction performance. These challenges are addressed through an SRH strategy that couples-controlled etching with Cl^- -mediated Br-I recrystallization. This synergistic process yields uniform surface potential, suppressed non-radiative recombination and high V_{oc} in single-junction UWBG devices with a -2.0-eV bandgap. When integrated with optimized MBG and NBG subcells, these improvements enable monolithic all-perovskite TJSCs to achieve 30.1% PCE (29.3% certified) and reasonable thermal, cycling and photostability. Collectively, our findings provide a viable strategy to address the long-standing challenges of UWBG perovskites in multijunction photovoltaics and offer a promising pathway towards the theoretical efficiency limit through further current matching and interfacial engineering.

Methods

Materials

All materials were used as received without further treatment. The organic halide salts (FAI, PEAI and CF₃-PACl) were purchased from Greatcell Solar Materials. PbBr₂ and CsI (99.9%) were purchased from Tokyo Chemical Industry. EDAl₂, 4F-PEACl and OAmCl were purchased from Xi'an Polymer Light Technology. Me-4PACz were purchased from TCI. PEDOT:PSS aqueous solution (AI 4083) was purchased from Heraeus Clevios. SnI₂ (99.999%) was purchased from Alfa Aesar. SnF₂ (99%), DMF (99.8% anhydrous), DMSO (99.9% anhydrous), ethyl acetate (99.8% anhydrous), anisole (99.8% anhydrous) and propan-2-ol (IPA, 99.8% anhydrous) were purchased from Sigma-Aldrich. HFIP (≥99.5%) and KOCN (97%) was purchased from Aladdin. C₆₀ was purchased from Nano-C. Tetrakis(dimethylamino) tin(IV) (99.9999%) was purchased

from Ai Mou Yuan Scientific Equipment. ITO-glass has a square resistance of 20 Ω, purchased from Suzhou ShangYang Solar Technology.

Perovskite precursor solution

UWBG FA_{0.7}Cs_{0.3}PbI₂Br₂ perovskite. A 0.9-M precursor solution of the UWBG perovskite (-2.0 eV) was prepared in a 4:1 (v/v) DMF/DMSO mixed solvent. The molar ratio of FAI/CsI/PbBr₂ was 0.7:0.3:1. PEAI was added to the precursor solution at a concentration of 1 mg ml⁻¹. KOCN was added to the precursor solution at an optimized concentration of 1 mg ml⁻¹ (refs. 1,35). After stirring at room temperature for 8 h, the precursor solution was filtered through a 0.22-μm PTFE membrane before use in film fabrication.

MBG FA_{0.8}Cs_{0.2}Pb(I_{0.9}Br_{0.1})₃ perovskite. A 1.5-M precursor solution of the MBG perovskite (-1.6 eV) was prepared in a 4:1 (v/v) DMF/DMSO mixed solvent. The molar ratio of FAI/CsI/PbI₂/PbBr₂ was 0.8:0.2:0.85:0.15. PEAI (1 mg ml⁻¹) was added to the precursor solution. After stirring at room temperature for 6 h, the precursor solution was filtered through a 0.22-μm PTFE membrane before use in film fabrication.

NBG FA_{0.7}MA_{0.3}Pb_{0.5}Sn_{0.5}I₃ perovskite. A 2.4-M precursor solution of the NBG perovskite (-1.22 eV) was prepared in a 2:1 (v/v) DMF/DMSO mixed solvent. The molar ratios of FAI/MAI and PbI₂/SnI₂ were 0.7:0.3 and 0.5:0.5, respectively. The precursor solution was prepared with a (FAI + MAI)/(PbI₂ + SnI₂) molar ratio of 1:1, to which SnF₂ was added at 10 mol% relative to SnI₂. Tin powder (5 mg ml⁻¹) and formamidine sulfonic acid (0.3 mol%) were introduced into the precursor solution to reduce Sn⁴⁺ and improve film uniformity. CF₃-PACl was introduced into the precursor solution as an additive at 0.3 mol%. After stirring at room temperature for 2 h, the precursor solution was filtered through a 0.22-μm PTFE membrane before use in film fabrication.

Single-junction UWBG solar cell fabrication

The pre-patterned ITO-glass substrates were cleaned sequentially with acetone and IPA. NiO nanocrystal (15 mg ml⁻¹ in water) was first spin-coated on ITO substrates at 4,000 r.p.m. for 10 s, followed by annealing at 130 °C for 30 min in air on a hotplate. 100 μl Me-4PACz solution (1 mmol in IPA) was dropped on an ITO substrate spinning at 4,000 r.p.m. (acceleration, 4,000 r.p.m. s⁻¹) for 10 s, followed by post-annealing at 100 °C for 5 min. To remove unbound molecules, the annealed film was subjected to a dynamic spin-casting process using 150 μl of IPA at 4,000 r.p.m. for 10 s. The washed film was then dried on a hotplate at 100 °C for 3 min. The cooled substrates were immediately moved into a nitrogen-filled glove box for perovskite deposition.

The perovskite films were deposited via a two-step spin-coating process: first at 1,000 r.p.m. for 10 s (ramp, 1,000 r.p.m. s⁻¹) and then at 4,000 r.p.m. for 40 s (ramp, 2,000 r.p.m. s⁻¹). Anisole (150 μl) was dispensed onto the spinning substrate 20 s before the completion of the second spin-coating step. To prepare UWBG perovskite films via the SRH strategy, the mixed polishing agent (HFIP/DMF = 30:1, v/v) containing OAmCl (1 mg ml⁻¹) was then spin-coated onto the wet perovskite film at 4000 r.p.m. for 40 s. The films were then annealed successively at 85 °C for 5 min and 135 °C for 15 min. Control films were fabricated using the same precursor and antisolvent procedure but without the subsequent SRH treatment.

For post-treatment of the UWBG perovskite, solutions of EDAl₂ and 4F-PEACl (each 1 mg ml⁻¹ in IPA) were prepared and deposited sequentially. Then, 100 μl of the treatment solution was spin-coated dynamically onto the perovskite film at 4,000 r.p.m. for 15 s. After each post-treatment solution was spin-coated, the perovskite films were annealed at 100 °C for 5 min. After cooling to room temperature, the substrates were transferred to an evaporation chamber for deposition of 26-nm C₆₀ at a rate of 0.2 Å s⁻¹. The substrates were then loaded into an atomic layer deposition (ALD) system (Veeco Savannah

S200) for deposition of 20-nm SnO_2 at 75 °C. The ALD- SnO_2 deposition conditions are detailed in our group's previous work³⁶. Finally, 150 nm of Cu was deposited as the back electrode by thermal evaporation. An anti-reflective coating was typically applied to minimize surface reflection.

All-perovskite tandem solar cell fabrication

The configuration of the TJSCs was ITO/NiO/Me-4PACz/UWBG perovskites/ $\text{EDAI}_2/4\text{F-PEACl/C}_{60}/\text{ALD-SnO}_2/\text{ITO/NiO/Me-4PACz/MBG perovskites/EDAI}_2/4\text{F-PEACl/C}_{60}/\text{ALD-SnO}_2/\text{Au/PEDOT:PSS/NBG perovskites/C}_{60}/\text{ALD-SnO}_2/\text{Cu}$. The UWBG solar cells were prepared as described above. After ALD- SnO_2 deposition, the substrate was then transferred to the radio frequency magnetron sputtering system to deposit 10-nm ITO. The sputtering pressure and power were 0.6 Pa and 50 W, respectively. The MBG perovskite cells were fabricated following the same procedures as the UWBG devices described above. The only differences lie in the NiO and perovskite layers. NiO nanocrystal (15 mg ml^{-1} in water/IPA, 3:1 v/v) was first spin-coated onto ITO substrates at 4,000 r.p.m. for 10 s and then annealed in air on a hotplate at 100 °C for 10 min. The perovskite films were deposited via a two-step spin-coating process: first at 2,000 r.p.m. for 10 s (ramp, 200 r.p.m. s^{-1}) and then at 6,000 r.p.m. for 30 s (ramp, 2,000 r.p.m. s^{-1}). During the second spin-coating step, 200 μl of anisole was dropped onto the spinning substrate 10 s before the end, and the substrates were then transferred to a hotplate and annealed at 100 °C for 10 min. After ALD- SnO_2 deposition, the substrates were transferred back to the thermal evaporation system for the deposition of an ultrathin Au cluster layer (~0.6 nm).

For the NBG perovskite cells, PEDOT:PSS layers were spin-coated in air onto the front cells and then annealed at 120 °C for 20 min in air. After cooling, the substrates were immediately transferred to a nitrogen-filled glove box for NBG perovskite film deposition. The perovskite films were deposited via a two-step spin-coating process: first at 1,000 r.p.m. for 10 s (ramp, 200 r.p.m. s^{-1}) and then at 4,000 r.p.m. for 40 s (ramp, 1,000 r.p.m. s^{-1}). During the second spin-coating step, 200 μl of ethyl acetate was dropped onto the spinning substrate 20 s before the end of the procedure. The substrates were then transferred to a hotplate and heated at 100 °C for 10 min. Then, 100 μl of EDAI_2 post-treatment solution (1 mg ml^{-1} in IPA) was dynamically spin-coated onto the perovskite film at 4,000 r.p.m. for 15 s. After cooling to room temperature, the substrates were transferred to the evaporation system for deposition of 20-nm C_{60} at a rate of 0.2 $\text{\AA s}^{-1}$. The substrates were then loaded into an ALD system (Veeco Savannah S200) for deposition of 20-nm SnO_2 at 75 °C. Finally, 150 nm of Cu was deposited by thermal evaporation as the back electrode, and an anti-reflective coating was typically applied to minimize surface reflection.

Characterization of solar cells

J - V characteristics of single-junction solar cells were measured with a Keithley 2450 source meter under a Class AAA solar simulator (Enlitech) at 100 mW cm^{-2} , calibrated against NREL-certified KG-5 and KG-0 reference cells for WBG and NBG devices, respectively. Unless otherwise noted, the J - V curves were recorded in a nitrogen-filled glove box at a scan rate of 10 mV s^{-1} (20-mV step, 10-ms delay). The active area was defined by aperture shadow masks (0.049 cm^2) placed in front of the solar cells. EQE spectra were measured in ambient air using a QE system (Enlitech) with monochromatic light focused on the device pixel at a chopper frequency of 20 Hz. For TJSCs, the J - V measurements were carried out under the illumination of a two-lamp high-spectral-match solar simulator (SAN-EI ELECTRIC, XHS-S0S1). EQE measurements were performed in ambient air, with bias light provided by LEDs for subcell-selective response. EQE measurements were performed in ambient air using bias light from bright LEDs. No bias voltage was applied during the EQE measurements of tandems. Bias light from LEDs was used with the following wavelength combinations: 700 + 850 nm

for the 2.00-eV UWBG subcell, 460 + 850 nm for the 1.60-eV MBG subcell and 460 + 700 nm for the 1.22-eV NBG subcell.

Stability tests of solar cells

A cover glass was bonded to the solar cells using UV epoxy (Three Bond) and cured under a 365-nm UV-LED lamp for 3 min.

MPP tracking. The operational stability tests were conducted under full AM1.5 G illumination (LED solar simulator, Guangzhou Cryscos Equipment) at an intensity of 100 mW cm^{-2} . The spectral irradiance of the LED simulator is provided as Supplementary Fig. 52. A custom-built LabVIEW-based MPP tracking system using a 'perturb and observe' algorithm was used under ambient conditions. The voltage at the MPP was determined from J - V measurements. The current density at a fixed bias voltage was monitored using a Keithley 2450 source meter. Ambient temperature was maintained at -25 °C, while the solar cell temperature rose to -35 °C under illumination due to the absence of active cooling during measurement.

Damp-heat test. Under open-circuit conditions, the cells were continuously stored in an environmental chamber at 85 ± 2 °C and $85 \pm 5\%$ RH for 240 h. After ageing, they were allowed to recover under open-circuit conditions at 23 ± 5 °C and RH of $\leq 75\%$ for 2–4 h, after which visual inspection and efficiency measurements were conducted.

Thermal cycling test. The fixtures used to secure the cells within the environmental test chamber ensured unrestricted air circulation. Temperature measurements were performed with an accuracy of ± 2.0 °C and a repeatability of ± 0.5 °C. The heating and cooling rates did not exceed 1.7 °C min^{-1} . The cells were subjected to 200 cycles between -40 °C and +85 °C, with a dwell time of at least 10 min at the maximum temperature. After ageing, visual inspection and efficiency measurements were conducted.

TOF-SIMS mapping measurements

TOF-SIMS was performed using TOF-SIMS M6 from IONTOF GmbH. Samples were analysed in the dual-beam profiling mode. All profiles were performed in the non-interlaced mode. Spectral data were acquired in the high-mass-resolution mode. The primary ion for analysis was 30-keV Bi^{3+} at 0.3 pA (with a Bi liquid metal ion source). This ion beam was applied over a $50 \times 50\text{-}\mu\text{m}^2$ area at the centre of the sputter crater. The sputter ion was a 10-keV Ar^+ cluster ion source with 2,500 Ar atoms. A $300 \times 300\text{-}\mu\text{m}^2$ sputter crater was used.

Femtosecond-resolved OPTP spectroscopy

Terahertz radiation was generated via optical rectification in a 0.5-mm-thick ZnTe (110) crystal, pumped by a commercial Ti:sapphire amplified laser (Libra, Coherent) with a 90-fs pulse duration, 400-nm centre wavelength and 1-kHz repetition rate. The radiation was detected by free-space electro-optic sampling in a 0.5-mm-thick ZnTe (110) crystal. Perovskite thin films on z-cut quartz substrates were photoexcited at 400 nm with fluences of 5–25 $\mu\text{J cm}^{-2}$. The terahertz amplitude change was monitored as a function of the time delay between the 400-nm pump and the terahertz probe using a translation stage. All measurements were performed in a nitrogen-purged environment to prevent degradation of the perovskite films in ambient air.

Other characterizations

Optical microscopy images were acquired using SIENTEC LT9000 in ambient air. KPFM measurements were performed on a Bruker Dimension Icon AFM under ambient conditions. A Pt-Ir-coated SCM-PIT probe was used in tapping mode, with a lock-in amplifier and signal filter configured for KPFM to achieve an electrical resolution of -10 mV. The conductive substrate of the perovskite samples was connected to the sample stage, and the scanning rate was set to 1 Hz to ensure image

quality. During the measurements, the AC bias voltage was applied to the tip, and the sample was grounded. Also, the AC bias produced an electric force on the cantilever, which is proportional to the potential difference between the conducting atomic force microscope tip and the sample. SEM imaging was performed using a TESCAN microscope at an accelerating voltage of 2 kV. XRD patterns were collected on a Bruker D8 Advance diffractometer with a NaI scintillation counter using monochromatized Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$). XPS analysis was performed on a Thermo Scientific Al K-Alpha XPS system. PL mapping was performed on a HORIBA LabRAM Odyssey with 532-nm excitation. Steady-state and time-resolved PL spectra were acquired using a HORIBA Fluorolog TCSPC system with PMT detectors. The samples were excited by a 485-nm pulsed laser at 1-sun equivalent intensity. Transient photovoltage decays were measured using a home-built system, where a 540-nm green LED modulated the V_{oc} under a constant light bias at a repetition rate of 2,000 Hz. The transient open-circuit voltage induced by the light perturbation was recorded by a digital oscilloscope with an input impedance of 1 M Ω . Femtosecond TA measurements were performed using a regeneratively amplified Yb:KGW laser (Light Conversion, PHAROS) at a 5-kHz repetition rate. The samples were prepared on glass substrates and translated at 1 mm s^{-1} during the measurement, with the pump fluence held at $4 \mu\text{J cm}^{-2}$. Sensitive EQE spectra were collected in ambient air using an FTPS PECT-600 system with a halogen lamp focused on the device pixel. The measurements were calibrated using Si and Ge reference cells. EIS measurements were carried out using a Solartron CHI660C impedance analyser over a frequency range of 10 Hz to 10 MHz at open-circuit voltage under dark conditions. The Nyquist plots and EIS parameters were fitted with ZView (version 4.2b; Scribner Associates). The $tDOS$ spectra were obtained with an Agilent E4980A precision LC meter.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data that support the findings of this study are available within the Article and its Supplementary Information. Additional data are available from the corresponding author on request. Source data are provided with this paper.

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Acknowledgements

We acknowledge the facilities and technical assistance provided by Nanjing University. We also acknowledge the Innovation Laboratory of Terahertz Biophysics at the National Innovation Institute of Defense Technology, the School of Engineering at The Australian National University and the PHI Analytical Laboratory at ULVAC-PHI Instruments Co., Ltd, for their valuable support.

Author contributions

H.T. conceived and directed the overall project. Y. Zhang and R.L. conceived the idea and designed the experiment. Y. Zhang, E.W., H. Liu and D.Z. fabricated all the devices and conducted the characterization. J.L. and C.C. performed OPTP measurements. A.D.B., K.N. and D.M. performed the PL mapping characterization. O.Y. and H.J. carried out the TOF-SIMS characterization. H. Li, D.X., H.Z., M.L., Y.W., C.D., Y. Zhu and L.L. helped with the device fabrication and material characterization. Y. Zhang and Y.L. organized the data and wrote the draft. Y.L. and H.T. reviewed and edited the paper. All authors contributed to the discussion and commented on the manuscript.

Funding

H.T. acknowledges financial support from the National Key R&D Program of China (grant number 2022YFB4200300), National Science Fund for Distinguished Young Scholars (grant number T2325016), Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China (grant number JYB2025XDXM410), Natural Science Foundation of Jiangsu Province (grant numbers BK20232022, BG2025024 and BK20233001), New Cornerstone Science Foundation through the XPLOER PRIZE, Program for Innovative Talents and Entrepreneur in Jiangsu, Fundamental Research Funds for the Central Universities (grant numbers 0213/14380206,

0205/14380252 and 0213/14380236). Y. Zhang acknowledges financial support from the National Natural Science Foundation of China (grant number 62404096), Natural Science Foundation of Jiangsu Province (grant number BK20241209) and Postdoctoral Fellowship Program of China Postdoctoral Science Foundation (grant number GZC20231092). R.L. acknowledges financial support from the National Natural Science Foundation of China (grant numbers 62305150 and 52427803). L.L. acknowledges financial support from the National Natural Science Foundation of China (grant number 62474086).

Competing interests

Patent (ZL202610056293.8) has been granted to Renshine Solar (Suzhou) Co., Ltd, based on this work, with Y. Zhang, H. Liu, R.L. and H.T. named as the inventors. H.T. is the founder, Chief Scientific Officer and Chairman of Renshine Solar (Suzhou) Co., Ltd, a company that is commercializing perovskite photovoltaics. The remaining authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41560-026-02135-1>.

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Peer review information *Nature Energy* thanks Philippe Holzhey, Yong-Young Noh and Atsushi Wakamiya for their contribution to the peer review of this work.

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Solar Cells Reporting Summary

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► Experimental design

Please check the following details are reported in the manuscript, and provide a brief description or explanation where applicable.

1. Dimensions

Area of the tested solar cells

☒ Yes
☐ No

The aperture area is 0.049 cm², which can be found in Methods section and main text. For independent verification, the aperture area is 0.0492 cm², which can be found in certification report (Supplementary Fig. 50).

Explain why this information is not reported/not relevant.

Method used to determine the device area

☒ Yes
☐ No

Detailed in Methods section, determined by aperture area. In certification report, determined by aperture area.

Explain why this information is not reported/not relevant.

2. Current-voltage characterization

Current density-voltage (J-V) plots in both forward and backward direction

☒ Yes
☐ No

Shown in main text and Supplementary information. Such Fig. 4b, Fig. 5c, Supplementary Fig. 39, Supplementary Fig. 46, Supplementary Fig. 48, Supplementary Fig. 51. Also showed in certification report (Supplementary Fig. 50).

Voltage scan conditions

☒ Yes
☐ No

Detailed in Methods section. For independent verification, which can be found in certification report (Supplementary Fig. 50).

Explain why this information is not reported/not relevant.

Test environment

☒ Yes
☐ No

The current-voltage characterization is performed in air at room temperature.

Explain why this information is not reported/not relevant.

Protocol for preconditioning of the device before its characterization

☐ Yes
☒ No

Provide a description of the protocol.

Detailed in Methods section. No precondition.

Stability of the J-V characteristic

☒ Yes
☐ No

Provided in main text and certified report. Such as Fig. 5g and Supplementary Fig. 50.

Explain why this information is not reported/not relevant.

3. Hysteresis or any other unusual behaviour

Description of the unusual behaviour observed during the characterization

☒ Yes
☐ No

Provided in main text and certified report. Such as Fig. 4b, Fig. 5c and Supplementary Fig. 50.

Explain why this information is not reported/not relevant.

Related experimental data

☒ Yes
☐ No

Provided in main text and certified report. Such as Fig. 4b, Fig. 5c and Supplementary Fig. 50.

Explain why this information is not reported/not relevant.

4. Efficiency

External quantum efficiency (EQE) or incident photons to current efficiency (IPCE)

☒ Yes
☐ No

Provided in main text, such as Fig. 4d, Fig. 5d.

Explain why this information is not reported/not relevant.

A comparison between the integrated response under the standard reference spectrum and the response measure under the simulator	☒ Yes ☐ No	The integrated short-circuit current density from the EQE spectrum matches the J_{sc} from the J-V measurements, see Fig. 4b, 4d, and Fig. 5c, 5d. Explain why this information is not reported/not relevant.
For tandem solar cells, the bias illumination and bias voltage used for each subcell	☒ Yes ☐ No	Detailed in Methods section. Explain why this information is not reported/not relevant.
 5. Calibration		
Light source and reference cell or sensor used for the characterization	☒ Yes ☐ No	Detailed in Methods section. Explain why this information is not reported/not relevant.
Confirmation that the reference cell was calibrated and certified	☒ Yes ☐ No	The reference cells were calibrated by NREL and explained in Methods section. Explain why this information is not reported/not relevant.
Calculation of spectral mismatch between the reference cell and the devices under test	☒ Yes ☐ No	The light spectrum used for measurements matches well with the reference silicon cell, and we did not calculate the spectral mismatch factor between the reference cell and the tested devices. Certified report with a spectra mismatch factor as described in Supplementary Fig. 50. Explain why this information is not reported/not relevant.
 6. Mask/aperture		
Size of the mask/aperture used during testing	☒ Yes ☐ No	Detailed in Methods section and and certification report (Supplementary Fig. 50). Explain why this information is not reported/not relevant.
Variation of the measured short-circuit current density with the mask/aperture area	☒ Yes ☐ No	In main text, we provided both the 0.049 cm ² apertures. Explain why this information is not reported/not relevant.
 7. Performance certification		
Identity of the independent certification laboratory that confirmed the photovoltaic performance	☒ Yes ☐ No	Certification report in Supplementary Fig. 50. Explain why this information is not reported/not relevant.
A copy of any certificate(s)	☒ Yes ☐ No	Certification report in Supplementary Fig. 50, including J-V, MPP. Explain why this information is not reported/not relevant.
 8. Statistics		
Number of solar cells tested	☒ Yes ☐ No	Provided in main text and Supplementary information, such as Fig. 4e and Supplementary Figs. 7, 16, 18, 20, 49. Explain why this information is not reported/not relevant.
Statistical analysis of the device performance	☒ Yes ☐ No	Provided in main text and Supplementary information, such as Fig. 4e and Supplementary Figs. 7, 16, 18, 20, 49. Explain why this information is not reported/not relevant.
 9. Long-term stability analysis		
Type of analysis, bias conditions and environmental conditions	☒ Yes ☐ No	Provided in main text (Figs. 5e-g). Explain why this information is not reported/not relevant.